

Design and Implementation of a Data Warehouse and Business Intelligence Solution for SuperStore Supermarket

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1. Introduction

1.1 Background of the Organization

SuperStore Supermarket is a supermarket chain in World that is managed by the John Keels group. The company has numerous branches spread in other parts of the country and has thousands of customers on a daily basis. SuperStore Supermarket sells very many products such as groceries, fresh products, household goods, personal care, technical, office products and packaged goods.

SuperStore Supermarket is a large organization in terms of retail, but it is associated with the large volumes of transactions per day in its branches. All the branches produce substantial operations and transactional data that may be exploited to enhance the performance and customer satisfaction of the business.

1.2 Business Problems Faced

Though SuperStore Supermarket gathers big amounts of data, this data may be gathered in different systems. This introduces a number of problems:

- As a problem with the analysis of sales in different branches.
- Lack of ability to find the best-selling products fast.
- Poor access to the customer purchasing trend.
- Lack of ability to compare monthly or annual sales patterns.
- Absence of management- decision integrated reporting.

Examples of questions that the management might be interested in knowing include:

- What type of product is the most profitable one?
- What is the least performing branch in terms of sales?
- What is the lowest month in terms of profit?
- What are the most valuable customers?
- How do products perform in terms of turnover?

In the absence of a centralized system, it is slow and inefficient to find the answers to these questions.

1.3 Need or a Data Warehouse

In order to overcome these challenges, SuperStore Supermarket must have a data warehouse that will bring together data of different sources into one repository that will be organized. A data warehouse enables:

- Centralized storage of historical data.
- Faster analytical queries.
- Multi- dimensional (product, branch, time, customer) analysis.

With the introduction of a data warehouse with the Galaxy Schema, SuperStore Supermarket will be able to study the sales trends and determine the most active regions, customer behavior, as well as streamline inventory management.

This will assist the management to make the decisions which are informed, drive up profitability, enhance customer satisfaction, as well as have competitive advantage in the retail market.

2. Identification of Data Sources

2.1 Data Sources Identification and Creation.

In order to design a good data warehouse to SuperStore Supermarket, data was taken out of a publicly available retail data depicting supermarket operations. The data set was modeled into various structured sources to model an actual organizational structure like CSV files, relational databases and Excel files.

This project utilized the following three main sources of data.

2.1.1 Data Source 1 – Customer Data (CSV File)

Source Type: CSV File

File Name: customer_data.csv

Description: The Orders dataset was used to extract the customer information and stored as a separate CSV file. This is a simulation of a customer registration or CRM (Customer Relationship Management) system that SuperStore Supermarket uses.

Structure of Customer Data:

Field Name	Data Type	Description
Customer_ID	Varchar	Unique identifier of any customer
Customer_Name	Varchar	Name of the customer
Segment	Varchar	Customer Segment
Country	Varchar	Country of residence
City	Varchar	City of residence
Region	Varchar	Geographical area
Market	Varchar	Market classification

Table 1: Structure of Customer Data

Use of Customer Data in Business Analysis:

This dataset supports:

- Determining high value customer groups.

- Geographical and market analysis of sales.
- Researching the buying behavior of customers.
- Carrying out customer segmentation analysis.

This information was utilized to fill the Dim_Customer table of the data warehouse.

2.1.2 Data Source 2 – Sales Transactions (SQL Database)

Source Type: Relational Database (MYSQL)

Table Name: sales_transactions.sql

Description: The Orders sheet was translated to MYSQL database in a transactional table. This emulates the Point-of-Sale (POS) terminal that is utilized throughout the branches of SuperStore Supermarket to track sales made daily.

This is the main source of data to construct Fact Sales table.

Table Structure of Transactions:

Field Name	Data Type	Description
Order_ID	Varchar	Unique sales order number
Order_Date	Varchar	Date on which order was placed
Ship_Date	Varchar	Date of order shipment
Customer_ID	Varchar	Refers to customer
Product_ID	Varchar	References product
Sales	Decimal	Total sales amount
Quantity	BigInt	Number of items sold
Profit	Decimal	Profit made on sale
Shipping_Cost	Decimal	Shipping cost charged
Shipping_Mode	Varchar	Delivery method
Order_Priority	Varchar	Urgency Level
Discount	Decimal	Discount put on the order

Table 2: Structure of Sales transaction Data

Use of Sales Transactions Data:

This dataset enables:

- Computation of revenue total.
- Discount impact analysis.
- Shipping cost evaluation.
- Determination of products that sell.

This operational information is the Fact Sales table of the Galaxy Schema.

2.1.3 Data Source 3 – Product Data (Excel File)

Source Type: Excel Spreadsheet

File Name: product_data.xlsx

Description: Orders data was converted into product information and saved in a different spreadsheet. This is used to simulate a product master file kept by the procurement or the inventory department at SuperStore Supermarket.

Structure of Product Data:

Field Name	Data Type	Description
Product_ID	Varchar	Product identifier is unique
Product_Name	Varchar	Product name
Category	Varchar	Product category
Sub_Category	Varchar	Sub_Category classification

Table 3: Structure of Product Data

The Product Data in business Analysis:

This dataset supports:

- Categorical and sub-category sales analysis.
- Comparison of performance of products.
- Recognition of most successful product groups.
- Profitability analysis on the basic of categories.

2.2 Importance of Multiple Data Sources

In actual organizations, such as SuperStore Supermarket, the data is maintained in various operation systems including CRM systems, POS systems, and inventory management systems. Such systems have been optimized to do routine day to day activities and not advanced analytical processing.

Using several sources to retrieve data and centralizing them in a data warehouse, the management can:

- Conduct thorough business intelligence.
- Generate strategic reports.
- Improve decision-making.
- Determine expansion opportunities.

These three sources of data are incorporated to form the basis on which the Galaxy Schema is designed as discussed in the following section.

3. Design the Data Warehouse Schema (Galaxy Schema)

3.1 Schema Design of Data Warehouse

In order to facilitate business intelligence and analytical reporting at SuperStore Supermarket, a Galaxy Schema is created.

A Galaxy Schema contains:

- Multiple Fact Tables
- Shared Dimension Tables

The reason why this would be an appropriate structure is that SuperStore Supermarket has several business processes that include:

- Sales Transactions
- Product Returns (based on transaction data)

The dimensions of both processes are common such as:

- Customer
- Product
- Location
- Shipping
- Date

3.2 Determination of Fact Tables

Fact tables are used to store quantifiable business information.

3.2.1 Fact Table 1 – Fact_Sales

All the sales are entered in this table.

Structure of Fact Sales

Field Name	Description
Sales_Key (PK)	Sale unique record
Order_ID	Original order number
Customer_Key (FK)	Related to Dim_Customer
Product_Key (FK)	Related with Dim_Product
Date_Key (FK)	Links to Dim_Date
Location_Key (FK)	Links to Dim_Location
Shipping_Key (FK)	Links to Dim-Shipping
Quantity	The number of sold items
Sales	Total revenue
Profit	Calculated profit
Shipping_Cost	Delivery cost
Discount	Discount applied
Shipping_Mode	Varchar
Order_Priority	Varchar

Table 4: Fact_Sales Table Structure

Such analysis can be done using this table:

- Calculation of total revenue and profit within all the branches.
- Sales trend analysis of sales monthly and annually.
- Assessment of the effects of discount on profit margins.
- Sales comparison on the regional and market level.
- Share of customers in total revenue.

3.2.2 Fact Table 2 – Fact_Returns

The returns are not explicitly represented in the dataset but can be determined by means of negative profit transactions or business rules.

This fact table stores refunded or returned transactions.

Structure of Fact_Returns

Field Name	Description
Return_ Key (PK)	Return record unique
Order_ID	Related order
Location_ Key (FK)	Related with Dim_Location
Date_ Key (FK)	Links to Dim_Date
Returned	Status of returned

Table 5: Fact_Returns Table Structure

This table supports:

- Determination of most returned products category.
- Revenue and profit loss measure, because of returns.
- Comparison of volumes of returns regionally.

3.3 Dimension Tables Identification

Business entities are defined in dimension tables.

3.3.1 Dim_Customer

Field Name	Description
Customer_Key (PK)	Surrogate Key
Customer_ID	Unique ID
Customer_Name	Customer name
Segment	Customer segment
Country	Country
City	City

Region	Region
Market	Market classification

Table 6: Dim_Customer Table Structure

Used for:

- Sales by segment
- Market comparison
- Customer behavior analysis

3.3.2 Dim_Product

Field Name	Description
Product_Key (PK)	Surrogate Key
Product_ID	Unique ID
Product_Name	Name of product
Category	Category of product
Sub_Category	Sub category

Table 7: Dim_Product Table Structure

Used for:

- Category performance
- Sub-category comparison
- Product profitability

3.3.3 Dim_Date

Field Name	Description
Date_Key (PK)	Unique ID
Full_Date	Actual date
Day	Day number
Month	Name of month
Month_Name	Full month name
Quarter	Q1, Q2, Q3, Q4
Year	Year
Day_of_Week	Day name
Is_Weekend	Does the date occur on weekend?

Table 8: Dim_Date Table Structure

Used for:

- Monthly trends
- Quarterly comparison
- Yearly growth analysis

3.3.4 Dim_Location

Field Name	Description
Location_Key (PK)	Unique ID
Country	Actual date
City	City
Region	Day number
Market	Name of month

Table 9: Dim_Location Table Structure

Used for:

- Comparison of sales performance of the branches and regions.
- Comparison of the regional returns.
- Subjective high and low performing markets.

3.3.5 Dim_Shipping

Field Name	Description
Shipping_Key (PK)	Unique ID
Ship_Mode	Delivery method
Order_Priority	Urgency Level

Table 10: Dim_Shipping Table Structure

Used for:

- Comparison of the cost of shipping by mode of ship.
- Shipping mode effect on the total profit margin.
- Determination of most used shipping methods by region.

3.4 Galaxy Schema

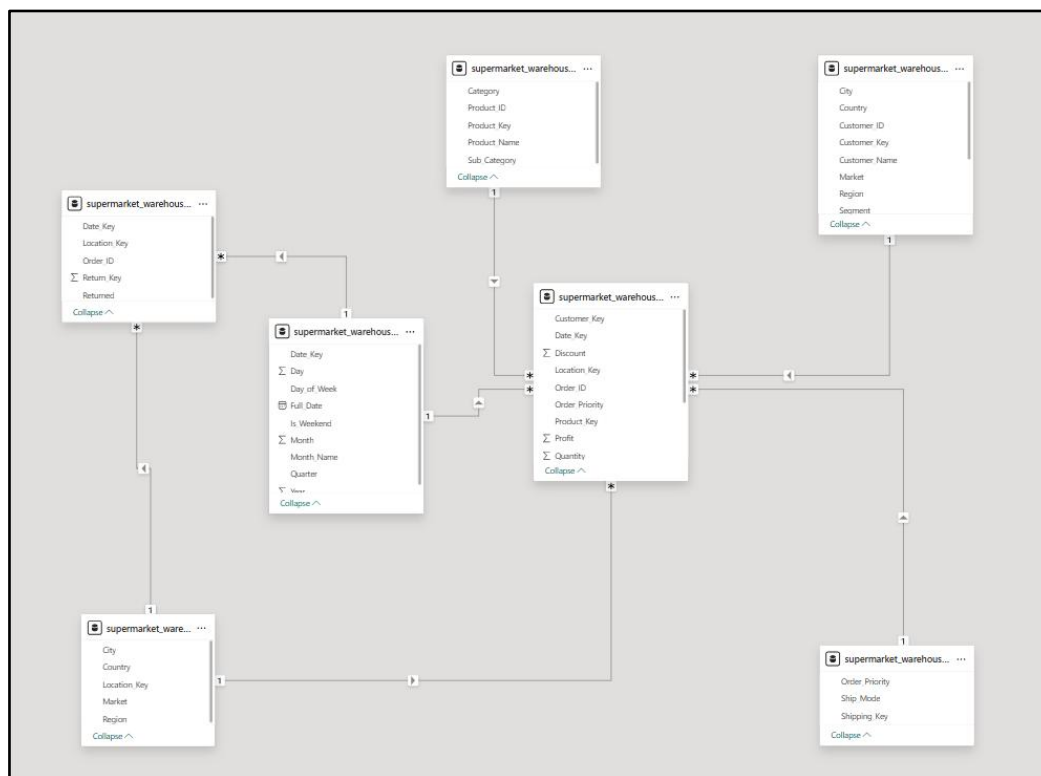
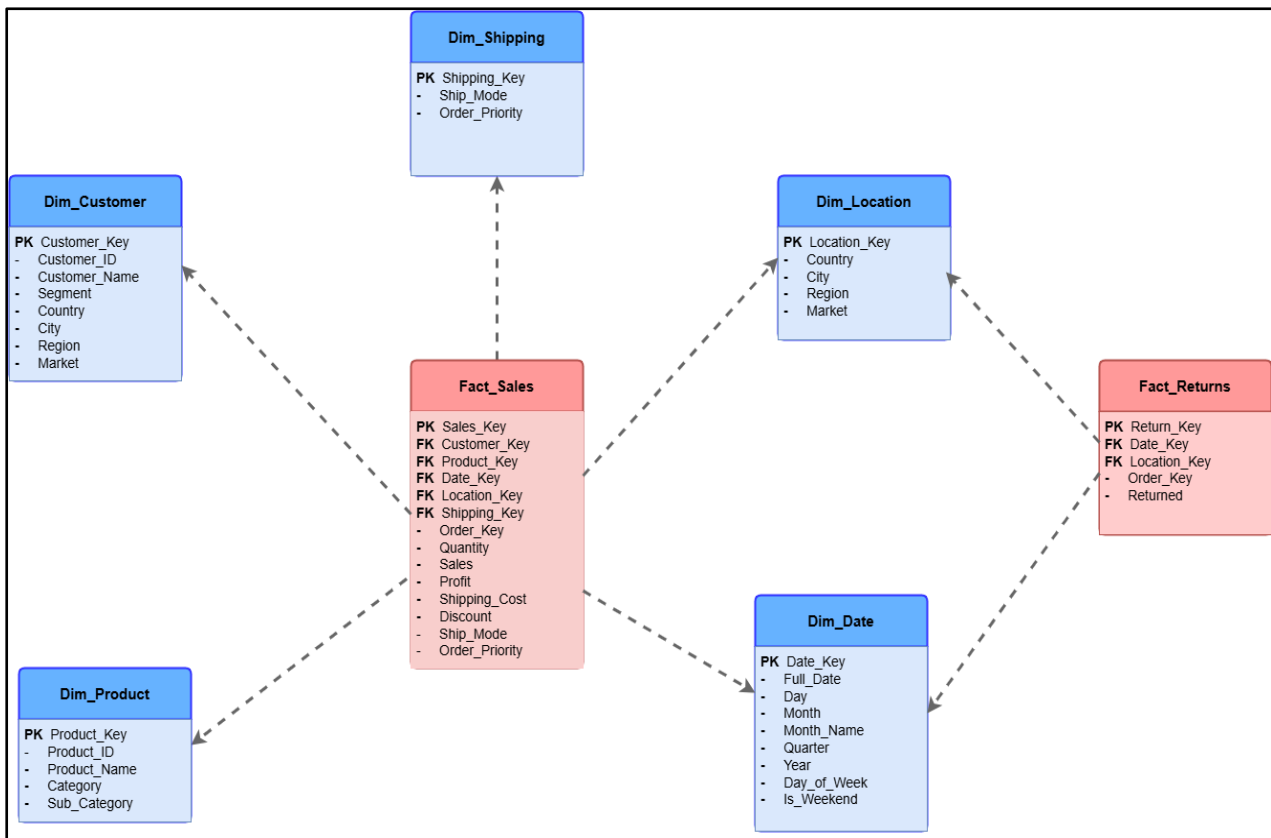


Figure 1: Galaxy Schema

3.5 Why Galaxy Schema is Suitable for SuperStore Supermarket

The Galaxy Schema is the best suitable data warehouse design that can be used by SuperStore Supermarket due to the following reasons:

- The SuperStore Supermarket has several business processes which are related, i.e., Sales and Returns each needed their own fact table.
- Dim_Date and Dim_Location are common dimensions in both FACTSALES and FACTRETURNS, which is the distinguishing feature of a Galaxy Schema.
- The schema is useful in the analysis of multi-dimensional analysis of various business processes simultaneously.
- It enhances query time to support high scale OLAP analytical workloads by normalizing shared dimension tables.

The Galaxy Schema allows the following OLAP operations, which the business intelligence at keels requires:

OLAP Operation	Description	Example at SuperStore Supermarket
Slice	Filter on a dimension value at a time	See all sales transactions in the year 2015 only
Dice	data across multiple dimensions at once	Sales in Asia Pacific in 2015Q3 Technology category
Drill-Down	Going to a more detailed view on a summary level	Going to quarterly then monthly to detailed daily sales
Roll-Up	Summarize detailed data to summary data	Individual city sales data to regional to global market sales data

Under this Galaxy Schema design, the SuperStore Supermarket Super management will be in a position to conduct the following business intelligence analyses:

- Comparison of sales revenue by region and year.
- Breakdown of contribution of profits by products and product sub-category.
- Return rate influence on total profitability per customer segment.
- Comparison of the shipping cost of various ship modes and market regions.

4. Physical Implementation of the Database Schema (Galaxy Schema Implementation in MYSQL)

4.1 Introduction

This section will give a physical implementation of the designed Galaxy Schema with MySQL.

This Galaxy Schema contains:

- Two Fact Tables:

- Fact_Sales
- Fact_Returns and
- Dimension Tables:
 - Dim_Customer
 - Dim_Product
 - Dim_Date
 - Dim_Location
 - Dim_Shipping



Figure 2: Successfully Created Tables

The reason of the creation of dimension tables first.

It is made since dimension tables are made first since the fact tables have a foreign key to the dimension tables. Establishing dimension tables ensures referential integrity and eliminates the constraint error when establishing fact tables.

4.2 Creating Dimension Tables

4.2.1 Dim_Customer

The descriptive data of customers is stored in the Dim_Customer table.

```
CREATE TABLE Dim_Customer (
  Customer_Key INT AUTO_INCREMENT PRIMARY KEY,
  Customer_ID VARCHAR(50),
  Customer_Name VARCHAR(100),
  Segment VARCHAR(50),
  Country VARCHAR(50),
```

Customer_Key	Customer_ID	Customer_Name	Segment	Country	City	Region	Market
1	AB-100151402	Aaron Bergman	Consumer	United States	Oklahoma City	Central US	USCA
2	JR-162107	Justin Ritter	Corporate	Australia	Wollongong	Oceania	Asia Pacific
3	CR-127307	Craig Reiter	Consumer	Australia	Brisbane	Oceania	Asia Pacific
4	KM-1637548	Katherine Murray	Home Office	Germany	Berlin	Western Europe	Europe
5	RH-9495111	Rick Hansen	Consumer	Senegal	Dakar	Western Africa	Africa
6	JM-156557	Jim Mitchum	Corporate	Australia	Sydney	Oceania	Asia Pacific
7	TS-2134092	Toby Swindell	Consumer	New Zealand	Porirua	Oceania	Asia Pacific
8	MB-1808592	Mick Brown	Consumer	New Zealand	Hamilton	Oceania	Asia Pacific
9	AB-100151404	Aaron Bergman	Consumer	United States	Seattle	Western US	USCA
10	AJ-107801	Anthony Jacobs	Corporate	Afghanistan	Kabul	Southern Asia	Asia Pacific
11	MM-7766110	MariaLuisa Morera	Consumer	Spain	Madrid	Western Asia	Asia Pacific

Figure 3: Created Dim_customer Table

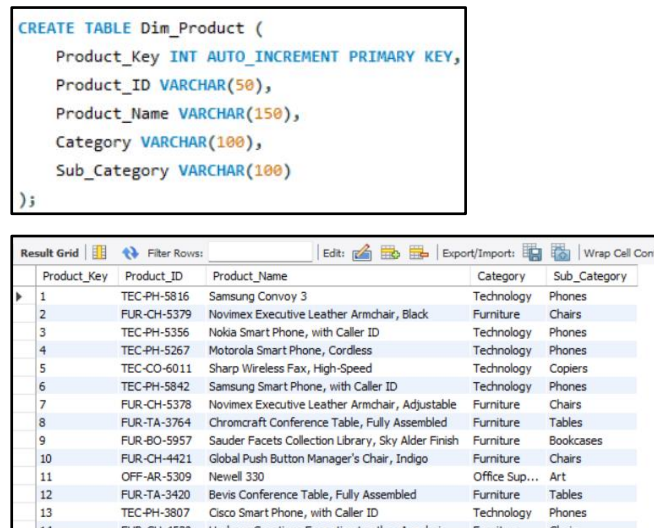
Purpose:

With the help of this table, it is possible to analyze customers by segmentation, to access the performance of a region, and to compare revenue received in the market.

4.2.2 Dim_Product

Product related descriptive attributes are stored in the Dim_Product table.

```
CREATE TABLE Dim_Product (  
    Product_Key INT AUTO_INCREMENT PRIMARY KEY,  
    Product_ID VARCHAR(50),  
    Product_Name VARCHAR(150),  
    Category VARCHAR(100),  
    Sub_Category VARCHAR(100)  
);
```



The screenshot displays a database interface with a 'Result Grid' showing 13 rows of product data. The columns are Product_Key, Product_ID, Product_Name, Category, and Sub_Category. The data includes various products like Samsung Convoys, Novimex armchairs, Nokia and Motorola phones, Sharp fax machines, Chromcraft conference tables, Sauder bookcases, and Newell office supplies.

Product_Key	Product_ID	Product_Name	Category	Sub_Category
1	TEC-PH-5816	Samsung Convoys 3	Technology	Phones
2	FUR-CH-5379	Novimex Executive Leather Armchair, Black	Furniture	Chairs
3	TEC-PH-5356	Nokia Smart Phone, with Caller ID	Technology	Phones
4	TEC-PH-5267	Motorola Smart Phone, Cordless	Technology	Phones
5	TEC-CO-6011	Sharp Wireless Fax, High-Speed	Technology	Copiers
6	TEC-PH-5842	Samsung Smart Phone, with Caller ID	Technology	Phones
7	FUR-CH-5378	Novimex Executive Leather Armchair, Adjustable	Furniture	Chairs
8	FUR-TA-3764	Chromcraft Conference Table, Fully Assembled	Furniture	Tables
9	FUR-BO-5957	Sauder Facets Collection Library, Sky Alder Finish	Furniture	Bookcases
10	FUR-CH-4421	Global Push Button Manager's Chair, Indigo	Furniture	Chairs
11	OFF-AR-5309	Newell 330	Office Sup...	Art
12	FUR-TA-3420	Bevis Conference Table, Fully Assembled	Furniture	Tables
13	TEC-PH-3807	Cisco Smart Phone, with Caller ID	Technology	Phones

Figure 5: Created Dim_Product Table

Purpose:

Facilitates the analysis of product category, sub-category and profitability analysis of the product.

4.2.3 Dim_Date

The Dim_Date offers time-based attributes of analysis.

```
CREATE TABLE Dim_Date (  
    Date_Key INT AUTO_INCREMENT PRIMARY KEY,  
    Full_Date DATE,  
    Day INT,  
    Month INT,  
    Month_Name VARCHAR(20),  
    Quarter VARCHAR(5),  
    Year INT,  
    Day_of_Week VARCHAR(20),  
    Is_Weekend BOOLEAN  
);
```

13	2014-06-06	6	6	June	05	2014	Friday	0
15	2013-11-13	13	11	November	04	2013	Wednesday	0
17	2013-12-30	30	12	December	04	2013	Wednesday	0
10	2013-04-10	10	4	April	05	2013	Friday	0
8	2013-03-08	8	3	March	01	2013	Tuesday	0
9	2013-04-14	14	4	April	05	2013	Friday	1
1	2013-11-06	6	11	November	04	2013	Tuesday	0
6	2014-06-30	30	6	June	05	2014	Friday	1
2	2014-11-02	2	11	November	04	2014	Wednesday	0
4	2014-01-30	30	1	January	01	2014	Tuesday	0
3	2014-10-13	13	10	October	04	2014	Friday	0
5	2014-05-02	2	5	May	01	2014	Wednesday	0
1	2014-11-11	11	11	November	04	2014	Tuesday	0
	Date_Key	Day	Month	Month_Name	Order	Year	Day_of_Week	Is_Weekend
	Year	Day	Month	Month_Name	Order	Year	Day_of_Week	Is_Weekend

Figure 6: Created Dim_Date Table

Purpose:

Facilitates monthly trends, quarterly comparison, annual growth analysis and weekend sales analysis.

4.2.4 Dim_Location

Geographical attributes are stored in the table called Dim_Location.

```
CREATE TABLE Dim_Location (
    Location_Key INT AUTO_INCREMENT PRIMARY KEY,
    Country VARCHAR(50),
    City VARCHAR(100),
    Region VARCHAR(50),
    Market VARCHAR(50)
);
```

Location_Key	Country	City	Region	Market
1	United States	Oklahoma City	Central US	USCA
2	Australia	Wollongong	Oceania	Asia Pacific
3	Australia	Brisbane	Oceania	Asia Pacific
4	Germany	Berlin	Western Europe	Europe
5	Senegal	Dakar	Western Africa	Africa
6	Australia	Sydney	Oceania	Asia Pacific
7	New Zealand	Porirua	Oceania	Asia Pacific
8	New Zealand	Hamilton	Oceania	Asia Pacific
9	United States	Seattle	Western US	USCA
10	Afghanistan	Kabul	Southern Asia	Asia Pacific
11	Saudi Arabia	Jizan	Western Asia	Asia Pacific
12	Brazil	Toledo	South America	LATAM
13	China	Mudanjiang	Eastern Asia	Asia Pacific
14	France	Paris	Western Europe	Europe

Figure 7: Created Dim_Location Table

Purpose:

Assists in regional sales comparison, local return analysis and location high-performing markets identification.

4.2.5 Dim_Shipping

Dim_Shipping table contains descriptive information of shipping.

```
CREATE TABLE Dim_Shipping (
    Shipping_Key INT AUTO_INCREMENT PRIMARY KEY,
    Ship_Mode VARCHAR(50),
    Order_Priority VARCHAR(50)
);
```

	Shipping_Key	Ship_Mode	Order_Priority
1	1	First Class	High
2	2	Second Class	Critical
3	3	First Class	Medium
4	4	Same Day	Critical
5	5	First Class	Critical
6	6	Standard Class	High
7	7	Standard Class	Low
8	8	Second Class	High
9	9	Standard Class	Medium
10	10	Second Class	Medium
11	11	Same Day	High
12	12	Same Day	Medium

Figure 8: Created Dim_Shipping Table

Purpose:

Enables shipping cost impact analysis, shipping mode profitability analysis and frequency of shipping mode.

4.3 Creating Fact Tables

4.3.1 Fact_Sales

Fact_Sales table contains quantifiable transaction information.

```
CREATE TABLE Fact_Sales (
    Sales_Key INT AUTO_INCREMENT PRIMARY KEY,
    Order_ID VARCHAR(50),
    Customer_Key INT,
    Product_Key INT,
    Date_Key INT,
    Location_Key INT,
    Shipping_Key INT,
    Quantity INT,
    Sales DECIMAL(12,2),
    Profit DECIMAL(12,2),
    Shipping_Cost DECIMAL(12,2),
    Discount DECIMAL(5,2),
    Order_Priority VARCHAR(50),
    Ship_Mode VARCHAR(50),

    FOREIGN KEY (Customer_Key) REFERENCES Dim_Customer(Customer_Key),
    FOREIGN KEY (Product_Key) REFERENCES Dim_Product(Product_Key),
    FOREIGN KEY (Date_Key) REFERENCES Dim_Date(Date_Key),
    FOREIGN KEY (Location_Key) REFERENCES Dim_Location(Location_Key),
    FOREIGN KEY (Shipping_Key) REFERENCES Dim_Shipping(Shipping_Key)
);
```

Result Grid												
Filter Rows:												
Edit: Export/Import: Wrap Cell Content: Fetch rows:												
	Sales_Key	Order_ID	Customer_Key	Product_Key	Date_Key	Location_Key	Shipping_Key	Quantity	Sales	Profit	Shipping_Cost	Discount
1		CA-2014-AB10015140-41954	1	1	1	1	1	2	221.98	62.15	40.77	0.00
2		MX-2014-CM1271582-41885	297	2	86	244	5	4	976.96	232.00	328.66	0.20
3		ID-2015-MM1726078-42181	675	2	535	151	9	6	2747.70	82.26	222.78	0.00
4		IN-2014-JR162107-41675	2	2	2	2	2	9	3709.40	-288.77	923.63	0.10
5		IN-2015-CR1282058-42294	184	3	180	156	6	5	3194.55	830.55	403.97	0.00
6		IN-2012-BG1174078-41108	179	3	175	151	1	4	2555.64	664.44	410.22	0.00
7		IN-2014-CS12175130-41788	240	3	232	100	10	6	3181.77	344.97	359.09	0.17
8		IN-2013-GP1474058-41419	47	3	50	45	2	3	1916.73	498.33	548.40	0.00
9		IN-2012-SC200507-41249	378	3	341	30	6	5	2875.10	511.10	302.01	0.10
10		IN-2014-CR127307-41929	3	3	3	3	3	9	5175.17	919.97	915.49	0.10
11		IN-2012-DL128657-40918	33	3	35	3	3	5	2875.10	511.10	665.27	0.10
12		IN-2013-CS1184527-41517	631	4	330	137	2	2	1285.56	51.42	232.17	0.00

Figure 9: Created Fact_Sales Table

The Supported Analytical Capabilities

This table enables:

- Computation of total revenue and profit.
- Trend analysis of the monthly and annual sales.
- Profit margins analysis under the impact of discounts.
- Customer analysis of overall revenue.

4.3.2 Fact_Returns

The Fact_Returns table captures the return or refunded transactions.

```
CREATE TABLE Fact_Returns (
    Return_Key INT AUTO_INCREMENT PRIMARY KEY,
    Order_ID VARCHAR(50),
    Date_Key INT,
    Location_Key INT,
    Returned VARCHAR(10),

    FOREIGN KEY (Date_Key) REFERENCES Dim_Date(Date_Key),
    FOREIGN KEY (Location_Key) REFERENCES Dim_Location(Location_Key)
);
```

	Return_Key	Order_ID	Date_Key	Location_Key	Returned
1		CA-2014-AB10015140-41954	1	1	Yes
2		CA-2014-AB10015140-41954	1	1	Yes
3		IN-2013-DL129257-41418	421	2	Yes
4		IN-2015-KM163757-42342	506	2	Yes

Figure 10: Created Fact_Returns Table

The Supported Analytical Capabilities

This table enables:

- Most categories of returned products were identified.
- Comparison of revenue and profit loss as a result of returns.
- Assessment of time series returns.

4.4 Explanation of relationship

The applied schema is a Galaxy Schema since:

- It has several fact tables (FactSales and FactReturns).
- The fact tables have common dimension tables.
- Referential integrity is guaranteed by foreign key relationships.
- Dimensions are used to offer descriptive background to numbers.

Every sales record is in reference to:

- One customer
- One product
- One date
- One location
- One shipping method

This structure can be used to do multidimensional time, geography, customers, and product analysis.

4.5 Why This Implementation is Effective

This analytical processing schema has a number of benefits:

- Data Integrity
 - Primary keys make the records in the dimension tables unique.
 - Foreign key constraints avoid the existence of orphan records in fact tables.
 - Referential integrity ensures that all the sales and returns refer to valid dimensions.
- Reduced Redundancy
 - Customer information is maintained in only one location in Dim_Customer rather than in each transaction.
 - Dim_Product contains product information that is centralized.
 - This gets rid of the update anomalies and provides consistency.
- Flexible Analysis
 - The users are able to analyze returns and the sales by combination of any dimensions.
 - Accommodates both extended transaction analysis and aggregate.
 - Addition of new attributes without restructuring is easy.
- Query Performance
 - Fine-tuning in support of aggregation queries in business intelligence.
 - For faster filtering, supports foreign key bitmap indexes.
 - Smaller table scans by dimensional filtering.

4.6 The way that the Schema Contributes to Business Intelligence.

This Structure:

- Improves query performance
- Favors the multidimensional analysis.
- Drill down and roll up are enabled.
- Efficiently integrates with BI tools e.g. Power BI.
- Enables trend, regional and profitability analysis.

5. ETL Process Implementation

5.1 Introduction

Extract, Transform and Load (ETL) is one of the basic operations in data warehousing that combines data of various sources of operations into a central system. The ETL pipeline will be used to guarantee the cleansing, transformation and organization of the raw data and its placement in the Data warehouse.

The ETL process as applied in this project was to blend customer, product and sales transaction information into Keels Data Warehouse that was designed based on Galaxy Schema.

ETL pipeline carries out several tasks as follows:

- Reading data on several source files.
- Populating the data on the staging tables.
- There is data cleaning and transforming.
- Inserting the transformed data in dimension tables.
- Inserting transactional information in fact tables.

The last warehouse tables are then linked to Power BI in order to have analytical reporting and visualization.

5.2 ETL Architecture

This project has employed the ETL architecture which has major layers.

5.2.1 Data Sources

There are three sources' files in which the data is obtained.

Data Source	Format	Description
Customer Data	CSV	Customer information.
Product Data	Excel	Product catalog
Sales Transactions	SQL database	Point of sale transaction information

5.2.2 Staging Layer

Raw data is then loaded onto staging tables before data is loaded into the warehouse.

The staging tables that have been developed are:

- Stagin_Customer
- Staging_Product
- Staging_Sales_Transactions

The staging layer serves as a working memory wherein raw data is checked and ready so as to be transformed.

5.2.3 Data Warehouse Layer

Dimension Tables:

- Dim_Customer
- Dim_Product
- Dim_Date
- Dim_Location
- Dim_Shipping

Fact Tables:

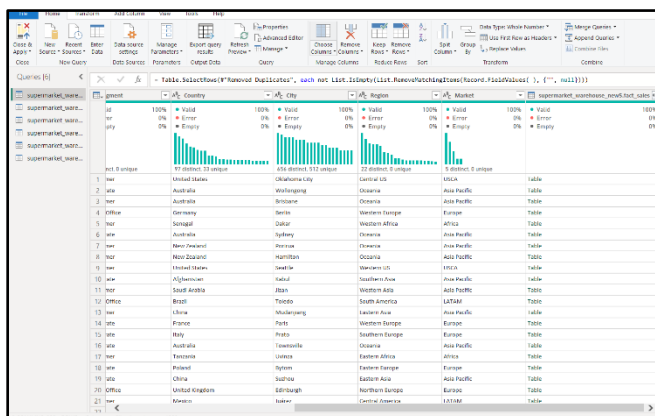
- Fact_Sales
- Fact>Returns

5.2.4 Business intelligence Layer.

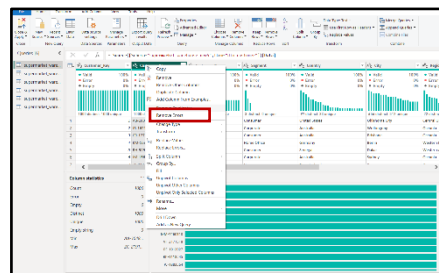
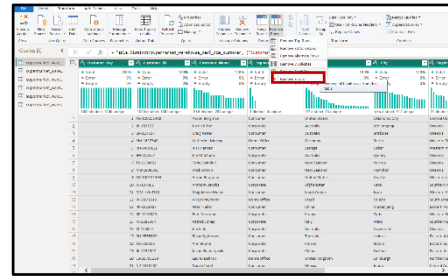
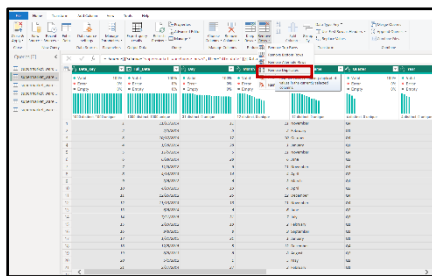
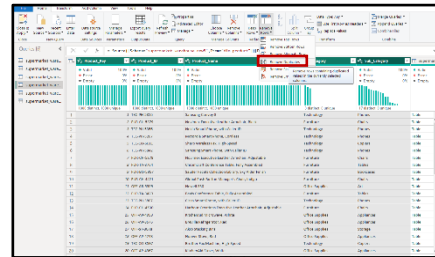
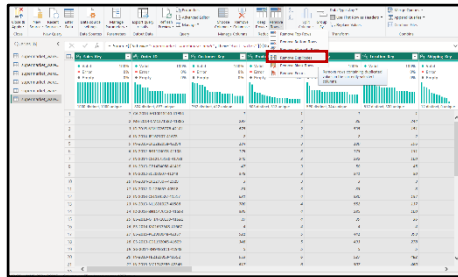
The warehouse tables are interconnected to the Power BI to carry out:

- Sales analysis Performance analysis of the products.
- Regional sales comparison
- The customer segmentation analysis.

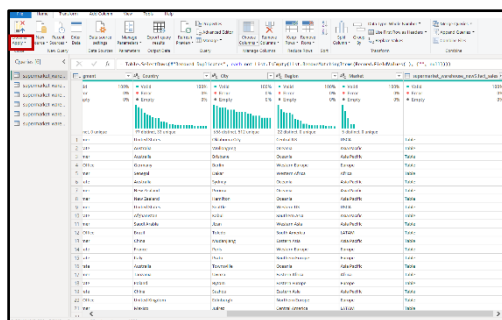
5.3 Data Extract



5.4 Data Transform



5.5 Data Load



Once it has been transformed, the cleaned data was added into the warehouse tables. Order of loading is considered significant since the fact tables are dependent on dimension tables.

Load of dimension tables was made initially:

- Dim_Customer

- Dim_Product
- Dim_Date
- Dim_Location
- Dim_Shipping

Load of fact tables was made initially:

- Fct_Sales
- Fact_Returns

5.6 ETL integration with Power BI

Once the process of ETL was complete, the data warehouse was linked to power BI.

The tables below were imported:

- Fact_Sales
- Fact_Returns
- Dim_Customer
- Dim_Product
- Dim_Date
- Dim_Location
- Dim_Shipping

The relationships are automatically identified with the help of the foreign keys of the schema, which is active in Power BI.

This integration facilitates business intelligence interactive dashboards.

5.7 Advantages of the ETL Implementation

The ETL that was implemented has a number of advantages:

- Improved Data Quality The cleaning and validation of data provide good analytical results.
- Data Integration Data in various systems get consolidated and centralized into one repository.
- Fact and dimension tables are optimized with an analytical query.
- Scalability ETL process can be expanded to accommodate other sources of data.
- Business Intelligence Support. It is easy to use the processed data to analyze it with tools like power BI.

6. Data Analysis and Visualization (Power BI)

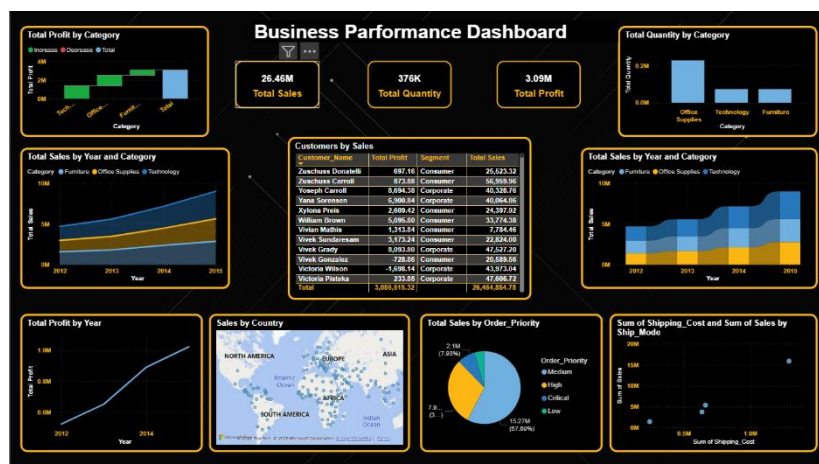
6.1 Introduction

Once the Galaxy Schema was implemented and the ETL process was done, the transformed data was loaded into Power BI to do an analytical reporting. This analysis aims to review the performance of sales, customer behavior, product trend, regional disturbing, shipping performance and the return pattern.

The Power BI dashboard gives a multidimensional perspective of the organization operation on the basis of combining data in fact and dimension tables. This facilitates strategic and fact-based decision making.

6.2 Graphical Visualization and Their Interpretation

Power BI visualization tools were used to analyze the data to define the patterns, trends and business insights with regards to the performance of the sales, product categories, the customers and shipping operation. Visualizations were made to investigate the data as follows.



1) Sales by Year and Category

- Types of visualization used: Stacked Area Chart
- Fields Used:
 - Axis – Year
 - Legend – Category
 - Values – Total sales

This is a visualization of the contribution made by various product categories to overall sales over several years. As seen in the chart, the sales have improved with time, reflecting the general growth of the business. Technology products are the most useful in terms of sales followed by Furniture and Office Supplies. The trend is that the demand of

the technology products is rising and it is propelling the growth in the sales of the company as a whole.

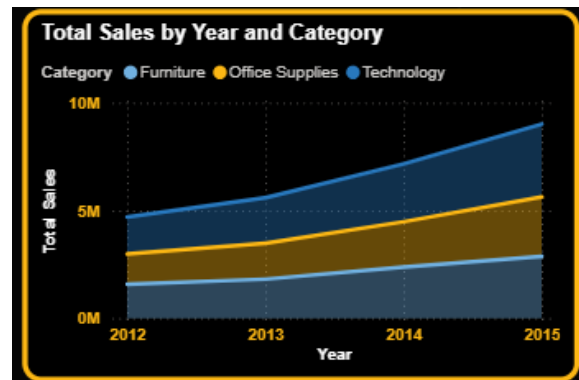


Figure 11: Visual 1

2) Categorical Ranking over Time

- Types of visualization used: Ribbon Chart
- Fields Used:
 - Axis – Year
 - Legend – Category
 - Values – Total sales

The ribbon chart shows the ranking of product categories in respect to sales in the various years. According to the visualization, it can be seen that Technology has always been in the top categories, whereas other categories have been changing ranking. This implies that technology products are highly demanded and marketable in the stable market, unlike other product categories.

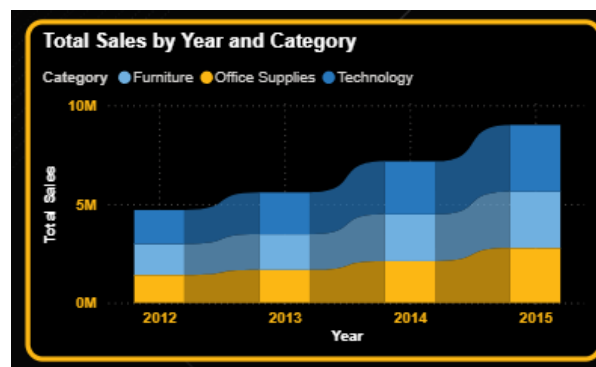


Figure 12: Visual 2

3) Profit Contribution by Category

- Types of visualization used: Waterfall Chart
- Fields Used:
 - Category - Category

- Values – Total Profit

The waterfall chart indicates the amount of contributions made by each category of products to the total profit. It puts emphasis on which categories are making positive contribution to profit, and which will contribute less to profit. The findings indicate that

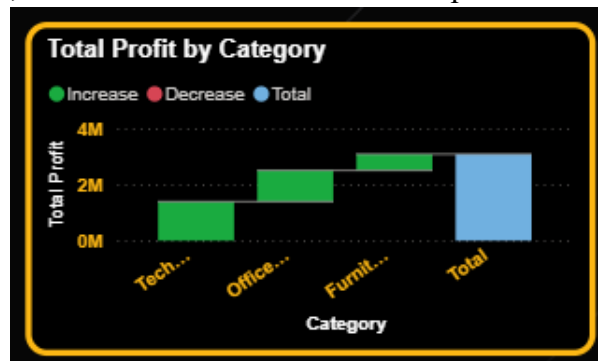


Figure 13: Visual 3

Technology is the most profitable with other categories carrying a lesser profit margin. This implies that the company ought to concentrate on marketing products of high-profit lineups.

4) Sales by Country

- Types of visualization used: Map
- Fields Used:
 - Location - Country
 - Values – Total Sales

This graph is the geographical distribution of sales in various countries. As shown in the map, there are certain countries that are making much more sales than others. Areas that are darker in color represent market performance that is stronger. This indicates that some countries are the core markets of the business and can provide the business a chance of further growth



Figure 14: Visual 4

5) Total Sales, Total Profit, Total Quantity

- Types of visualization used: Card
- Fields Used:
 - Values – Total Sales, Total Profit, Total Quantity

This KPI card shows the number of profits, sales and quantity earned out of all sales transactions. The comparison of the total profit, total quantity and the total sales assists in the determination of the profitability of the business. A good profit amount means that the organization is making money out of its business.

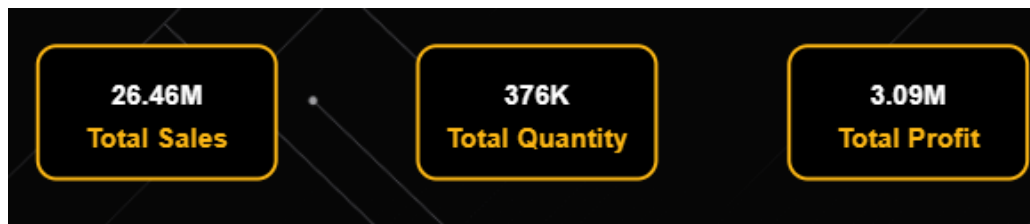


Figure 15: Visual 5

6) The Number of Items Sold by Category

- Types of visualization used: Stacked Area Chart
- Fields Used:
 - Axis – Category
 - Values – Total Quantity

This chart will give a comparison of the items that have been sold in the various categories of products. The findings indicate that some of the categories are selling very high volumes of sales compared to others. This trend can be used to inform inventory planning and marketing policy as it can indicate what product type is the most popular with customers.

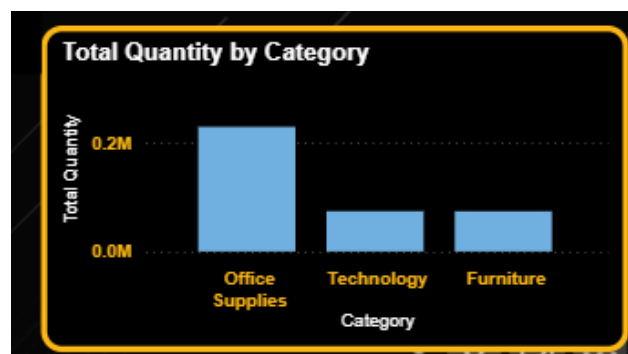


Figure 16: Visual 6

7) Distribution of Priority in an Order

- Types of visualization used: Pie Chart
- Fields Used:
 - Legend – Order_Priority
 - Values – Total Sales

The pie chart demonstrates the percentage of sales according to the level of order priorities. The allocation shows how sales are apportioned in various categories of priorities. A higher percentage at specified levels of priority implies that the majority of orders belong to definite operational groups and they assist the business to realize the patterns of order handling.

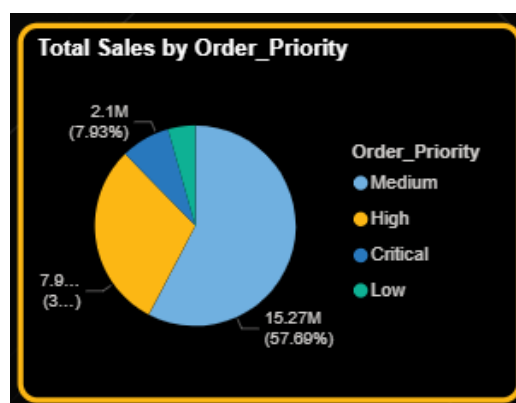


Figure 17: Visual 7

8) Sales VS Shipping Cost

- Types of visualization used: Scatter Chart
- Fields Used:
 - X Axis – Shipping_Cost
 - Legend – Ship_Mode
 - Y Axis – Sales

The scatter chart examines the correlation between the cost of shipping and its sales value. The trend indicates that orders that have greater sales value are likely to be expensive in terms of shipping. Various shipping options are also observable and one is



Figure 18: Visual 8

able to compare delivery modes. This observation can be used to realize the relationship between shipping cost and order size and logistical choices.

9) Profit Trend Year-by-Year

- Types of visualization used: Line Chart
- Fields Used:
 - Axis – Year
 - Legend – Total Profit

The line chart will indicate the trend of the profit of the company over the years. Trends of increasing denote better performance of business and augmented profitability whereas reducing trends may depict transitions in sales performance or cost of operation. The following visualization is useful in identifying the general increase in profits and annual variations.

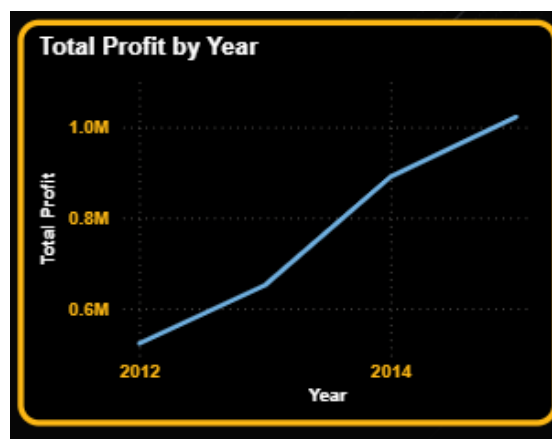


Figure 19: Visual 9

10) Customers by Sales

- Types of visualization used: Table
- Fields Used:
 - Columns – Customer_Name, Total Sale, Total Profit, Segment

The table presents the best customers in terms of total sales and profit. It will be useful in determining high value customers who generate high revenue to the company. It is by knowing these customers that the business is able to concentrate on customer retention and target marketing.

Customer_Name	Total Profit	Segment	Total Sales
Zuschuss Donatelli	697.16	Consumer	25,523.32
Zuschuss Carroll	873.88	Consumer	56,959.96
Yoseph Carroll	8,694.38	Corporate	40,328.76
Yana Sorensen	6,900.84	Corporate	40,064.86
Xylona Preis	2,609.42	Consumer	24,397.92
William Brown	5,095.80	Consumer	33,774.38
Vivian Mathis	1,313.84	Consumer	7,784.46
Vivek Sundaresam	3,173.24	Consumer	22,824.00
Vivek Grady	8,093.80	Corporate	47,527.20
Vivek Gonzalez	-728.86	Consumer	20,589.56
Victoria Wilson	-1,698.14	Corporate	43,973.04
Victoria Pisteka	233.88	Corporate	47,606.72
Total	3,086,515.32		26,464,854.78

Figure 20: Visual 10

6.3 Key Patterns Observed

Based on the analysis, a number of key trends were determined:

- The trend is revenues and profits are seasonal.
- There are product categories that generate a lot of profit.
- Certain groups of customers bring more income.
- There is a great difference in regional performance.
- The mode shipping determines the cost of operation.
- The returns are concentrated in specific areas or types of products.

These observations indicate the significance of performance monitoring based on data.

6.4 Business Intelligence Deduced

The Power BI dashboard would allow the management to:

- Real-time sales and profit monitoring.
- Determine products with potential gains.
- Identify expensive shipping strategies.
- Assess performance on a regional basis.
- Evaluate customer segment and minimize losses.

The strategic planning and competitive advantage are improved due to the integration of data warehousing and business intelligence tools.

7. Data Ethics

7.1 Introduction to Data Ethics

Data Ethics are defined as the responsible gathering, storage, processing and analysis of data in a way that allows guaranteeing privacy, security, equity, transparency and legality. The data warehouse that is implemented in this project combines transactions, customer, product, location and shipping information. This information is useful to make strategic decisions and therefore it needs to be treated in a moral manner to safeguard the stakeholders and uphold integrity within the organization.

7.2 Data Privacy and Confidentiality of Customers

The Dim_Customer table is the table that holds attributes associated with customer like:

- Customer ID
- Customer Name
- Segment
- Country
- City
- Region
- Market

Even though such data is utilized in an analytical mode, data that can be identified by customers should be secured. The privacy is violated and the reputation can also be damaged due to unauthorized access to personal data.

To ensure confidentiality:

- Only authorized human beings should access the data warehouse.
- Sensitive areas like names of customers should not be reported in public.
- External reports are supposed to be done using aggregate information rather than individual information.

Ensuring privacy of customers enhances trust and promotes ethical business conduct.

7.3 Data Security and Confidentiality

There are several fact and dimension tables that are integrated in data warehouse such as sales, return, product and shipping information. This is a centralized storage that predisposes it is an important business asset.

The security measures should entail:

- Access to the database is secured by passwords.
- Network security and firewall.
- Existing connection between MYSQL and Power BI.
- Regular database backups.

These measures are used in avoiding data breach, unauthorized access and loss of data.

7.4 Data Accuracy and Integrity

Business intelligence needs accurate data. Because of the nature of the ETL process where data is extracted and transformed, data quality is very essential.

The following are some of the measures taken to ensure the integrity of data:

- Defining referential integrity by defining primary and foreign keys.
- Authenticating numbers like Sales, Profit and Shipping Cost.
- Removing duplicate records.

Proper and consistent data in use will avoid misguided analysis and wrong managerial decision making.

7.5 Making the Best Use of Analytical Wisdom Ethically.

The Power BI dashboard breaks down sales by customer segment, region, product and shipping mode. Although these insights are important in enhancing the performance of a business, they should be applied with caution.

For example:

- Some segments of customers should not be unfairly discriminated against by marketing strategies.
- The data of performance in the region must not be abused by sidelining certain markets.
- The analysis of returns should be directed towards better quality of products instead of faulting particular areas.

Ethical analysis is one that ensures fairness and responsible decision making in business.

7.6 Responsible Handling of Return and Performance

Data are designed to guarantee that the management has efficiently handled its return and performance data. The Fact_Returns table determines the pattern of returns in regions and products. Though this information is valuable in defining the quality problems, it should be used cautiously.

Misinterpretation may:

- Cause inequitable supplier selection.
- Fraudulently represent customer behavior.
- Develop discriminatory working policies.

Consequently, the decisions are required to be made relying on confirmed trends and with the supplementary contextual analysis.

7.7 Legal and Regulatory Compliance

The company must ensure legal and regulatory compliance in all countries where it operates. Organizations should adhere to the relevant data protection and privacy laws. The collection and utilization of customer and transactional data must be done on justifiable business purposes.

The data warehouse must:

- Restrict the attributes of data collection.
- Do not keep too much personal information.

Conformity to the law safeguards the organization and the consumers.

7.8 Importance of Ethical Data Practices

As ethical data practice in the data warehouse will guarantee:

- Customer privacy protection.
- Safe management of company information.
- Increase organizational image.
- Credible and objective analytical ideas.

Business intelligence implementation and data warehousing therefore requires ethical considerations to be one of its core elements.

8. Conclusion

The project was able to design and install a full Data Warehouse solution based on a Galaxy Schema model. The system unites the transactional sales information, the return and the customer information, the product information, the location information, and shipping information to a model analytical setting.

The implementation process entailed:

- Determination of various data sources.
- With using ETL techniques, doing data cleaning and transformation.
- Creation of dimension and fact tables.
- Setting up of primary and foreign key relationship.
- Having referential integrity constraints.
- Creation of an interactive Power BI dashboard.

The Galaxy Schema design allowed performing the multidimensional analysis based on various perspectives of the business, such as time, product categories, customer segments, geographic regions and shipping methods.

The relevant insights were created with the help of the Power BI dashboard, including:

- High-performing products category identification.
- Identification of seasonal sales.
- Profitability analysis at the sub-categorizations.
- Competition of the regions.
- Cost impact analysis of shipping.

These lessons prove the benefits of structured data warehousing on strategic decision-making, operations and competitive advantage.

Moreover, the moral aspect was considered to provide responsible treatment of the data of the customers and transactions. The privacy of data, its security, integrity, and impartial analysis were also raised as critical elements of the data warehouse implementation.

To sum up, this project defines the significance of using data warehousing and business intelligence applications in an attempt to convert raw data into useful organizational knowledge. The deployed system is scalable, reliable and efficient analytical framework enabling the use of data to make decisions.

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